Nadir ZOUAOUI

Electrical Engineer

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Class B driver's license | Personal vehicle | National service: exempt | nadirzouaoui.github.io/projects

PROFESSIONAL SUMMARY

State engineer in electrical engineering with four years of multidisciplinary experience across Oil & Gas, industrial automation and renewable energy. Covers the full project cycle, from design and engineering to wiring, commissioning and site work. Motivated, dynamic and passionate about electrical engineering, with the goal of applying my skills to innovation and sustainable development.

CORE SKILLS

Oil & Gas | Industrial Equipment | Hazardous Areas (ATEX) | MCC & Switchgear | Commissioning | Photovoltaic system design and sizing | Electrical diagrams and schematics | SCADA | Industrial automation

TECHNICAL TOOLS

EPLAN Electric P8 + Pro Panel | AutoCAD Electrical | Caneco BT | Archelios PV simulation | TIA Portal, STEP 7 | Siemens S7-1200 | Microsoft Excel | Blender 3D | Adobe Suite

LANGUAGES

Arabic - Native | French - Bilingual, C1 certified (DALF 76/100) | English - Fluent

CERTIFICATIONS

Electrical Equipment in Hazardous Areas (EEHA) - ROAR Training Solutions (Australian RTO)

2026 | Online | Australian national competency units (AS/NZS 60079, aligned with IEC 60079)

- Hazardous-area classification, explosion-protection requirements, planning and installation of Ex equipment and wiring (UEEHA0004, 0022, 0029, 0025).
- Detailed, visual and close inspection, maintenance and periodic inspection programs for hazardous areas (UEEHA0020, 0038, 0026, 0023; UEECD0011).

Low Voltage Design Fundamentals and Caneco BT Training - Lokelect Engineering

September 2026 | 40 hours | Online | Certificate ID 2b3376a9aea18854

- LV design: earthing systems, power balance, cable sizing, voltage drop, selectivity basics.
- Caneco BT: project setup, modeling, results analysis and design iteration.

PROFESSIONAL EXPERIENCE

Electrical Engineer - RMBTECH SARL

March 2025 - Present | Oran, Algeria

- Industrial rehabilitation: technical direction, electrical redesign (EPLAN), rewiring and automated commissioning (Siemens S7-1200) of a BOSSAR packaging machine for GBFoods (Jumbo).
- SCADA gas pipeline projects (Sonatrach): led electrical engineering, including a site audit of 66 isolated stations (solar/grid powered) on the GK3 gas pipeline, deployment on GK1/GK2, and pre-engineering of GR1/2/4/6 (integrator: SUPCON).
- Critical equipment and ATEX: engineering for the replacement of an 8500 A rectifier for the AOA electrolysis system, and design of an ATEX Ex "p" pressurized cabinet for Sonatrach GP2Z (Arzew).
- Agro-industry: electrical design for the SCADA system of the OAIC grain silo in Mesra.

Lead Electrical Engineer - PROSKID EURL

July 2023 - March 2025 | Algiers, Algeria

- Designed and sized electrical systems of Oil & Gas skids for clients ENGCB/GCB and SARPI.
- Produced power supply, power and control schematics in EPLAN Electric P8 and AutoCAD Electrical.
- Designed and sized the off-grid solar power supply systems feeding the skids.
- Commissioned 27 Milton Roy chemical anticorrosion dosing skids and their solar installations.
- Designed and built an ATEX Ex "d" flameproof enclosure for a corrosion inhibitor skid, including component mounting and wiring.

Photovoltaic Design Engineer - Sungy SPA

October 2022 - July 2023 | Algiers, Algeria

- Delivered over 300 kWp of installed solar capacity.
- Formulation of energy sobriety plans and solutions.
- 3D layout, sizing and simulation of PV systems.
- Electrical sizing and drafting of electrical diagrams for solar installations.
- Commissioning and monitoring of solar installations.
- Design of deployable mobile solar solutions.

Document Controller - CEVIELEC SPA (Cevital Electrique)

March 2022 - October 2022 | Illizi, Algeria

- ALRAR Gas Processing Complex re-instrumentation project.
- Management of revisions, modifications and document markups.
- Project planning and tracking.
- Cabinet wiring and instrument installation.
- Loop checks and troubleshooting.

FREELANCE

5.4 MW Industrial Solar Installation - SARL Ibrahim & Fils (IFRI)

Akbou, Bejaia, Algeria

- Rooftop PV design for a zero-injection self-consumption plant over several production sites: JinkoSolar 635 Wc bifacial modules, Huawei SUN2000-150K inverters, 10 x 241 kWh battery storage.
- Layout, string and inverter zoning, and DC/AC cable sizing optimized for voltage drop and cost.
- Single-line diagram dossiers and load and power quality analysis.

Electrical Training Modules - ROAR Training Solutions (Australian RTO)

- Developed first-person 3D playable exercises for electrical competency training, including a 3D substation exercise and a Low Voltage Rescue and CPR simulation.
- Rescue scenario: hazard identification, PPE, breaker isolation, insulated-hook rescue and casualty drag, then CPR with AED, guided by a phase-gated procedure checklist.
- 3D assets and character animation in Blender; interaction logic and training scenarios in Godot 4.
- Packaged as SCORM 1.2 modules running in the browser (WebGL2) for delivery through the client's LMS.

Motion Design & Video

- Freelance motion design and video work (Behance: nad04799c26; Freelancer.com/u/NadirZouaoui; Fiverr: nadir_z).

EDUCATION

State Engineer + Master II, Electrical Engineering

Ecole Nationale Polytechnique d'Oran - Maurice Audin (ENPO-MA), Oran, Algeria | 2018 - 2021

- Valedictorian, class of 2018-2021.

Preparatory Cycle in Sciences and Technologies

Ecole Nationale Polytechnique d'Oran - Maurice Audin (ENPO-MA), Oran, Algeria | 2016 - 2018

PROJECTS

Design and Realization of a 5 kW Autonomous Photovoltaic Microgrid - Master's Thesis

May - July 2021

- Sizing of a microgrid comprising a 3 kW PV array, an 800 Ah battery bank, a 50 kVA diesel generator and a hybrid inverter supporting three input sources (PV, battery, grid/genset).
- Design of electrical and synoptic diagrams.
- Realization of outdoor (PV array and mounting structure) and indoor (inverter, batteries, generator, electrical boxes, ATS) installations.
- Commissioning and testing of the microgrid.

INTERNSHIPS

SKTM - Dhaya Production Unit

March - April 2021

- End-of-studies internship at the 12 MW photovoltaic power plant of Dhaya, Telagh, Sidi Bel Abbes.
- Study of large-scale photovoltaic production systems and overview of the SCADA system.

SDO Sonelgaz

December 2019

- Study of the MV-LV distribution network of Sidi Bel Abbes.
- Visits to HV-MV and MV-LV transformer stations and substations.

ENIE Solar

April 2019

- Internship in the photovoltaic panel production unit in Sidi Bel Abbes.
- Overview of the PV panel manufacturing process; troubleshooting and machine maintenance.

CITAL SPA

December 2018

- Overview of energy transmission and traction systems of the Sidi Bel Abbes tramway.
- HVDC overhead transmission line and AC/DC substations; electric traction with asynchronous motors.